

Percentages (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)	20	1	
1(b)	90	2	M1 for $600 \times \frac{5}{100} \times 3$
1(c)	60	2	M1 for $90 \div 1.5$ or $90 \div \frac{150}{100}$
2(a)	605 grams	3	M2 for $500 \times \left(\frac{110}{100}\right)^2$ or $500 \times 1.1 \times 1.1$ or M1 for $500 \times 1.1 = 550$
2(b)	121%	1	B1 for 121% or their $\frac{605}{500} \times 100$
3(a)	280 GBP	2	M1 for $250 \times \frac{112}{100}$ or $250 + 30$
3(b)	252 GBP	2	M1 for their $280 \times \frac{90}{100}$ or their $280 - 28$
4(a)	150 EUR	3	M2 for $180 \div \frac{120}{100}$ or $180 \times \frac{100}{120}$ or M1 for identifying $120\% = 180$
5(a)	40	2	M1 for $\frac{210 - 150}{150} \times 100$
5(b)	5400	2	M1 for $5000 \times 1.2 \times 0.9$
5(c)	250	2	M1 for $\frac{150}{60} \times 100$
6(a)	5	2	M1 for $75 = \frac{500 \times 3 \times T}{100}$
6(b)	2000	2	M1 for $2800 \div 1.4$ or $\frac{2800}{1.4}$
6(c)	306	2	M1 for $400 \times 0.85 \times 0.9$
7(a)	16%	4	M1 for assigning a base value e.g., 100 stu- dents (60 girls, 40 boys) M1 for calculating new boys $40 \times 1.25 = 50$ M1 for calculating new girls $60 \times 1.1 = 66$ A1 for new total 116 leading to 16%
8(a)	Correct proof	2	M1 for 1.5×0.8 or $1.5x - 0.2 \times 1.5x$ A1 for 1.2 leading to 20%
8(b)	125	2	M1 for $0.2x = 25$ or $\frac{20}{100}x = 25$

Part	Answer	Marks	Partial Marks / Guidance
9(a)	Correct proof	2	M1 for $\frac{P \times r \times 2}{100} = 120$
9(b)	$r = 5, P = 1200$	4	M1 for $P \left(1 + \frac{r}{100}\right)^2 - P = 123$ M1 for $P \left(\frac{2r}{100} + \frac{r^2}{10000}\right) = 123$ M1 for $120 + \frac{6000r}{10000} = 123$
10(a)	1039.50 GBP	2	M1 for $4500 \times 0.0385 \times 6$
10(b)	5539.50 GBP	1	FT for $4500 + \text{their (a)}$
11(a)	555000 or 555193	3	M2 for 452000×1.042^5 or M1 for 452000×1.042
11(b)(i)	5.8% increase	3	M2 for $1.15 \times 0.92 = 1.058$ or M1 for either 1.15 or 0.92 seen
11(b)(ii)	Let original area be $A = L \times W$. New area = $1.15L \times 0.92W$.	2	M1 for correct algebraic setup.
	$= 1.058 \times L \times W = 1.058A$.		A1 for final correct statement with no errors.
11(c)	283000 or 283111	3	M2 for $318500 \div 1.125$ or M1 for $112.5\% = 318500$
12(a)	14 300	3	M2 for $24500 \times (1 - 0.125)^4$ or M1 for 24500×0.875
12(b)	994.12	3	M2 for $845 \div 0.85$ or M1 for 85% associated with 845
12(c)	8	3	M2 for $1.052^n > 1.5$ (or trial with 8 and 7) M1 for $8000 \times 1.052^n = 12000$
13(a)	9.66	2	M1 for $\frac{1450000 - 1310000}{1450000} \times 100$
13(b)	1 960	3	M2 for $2450 \div 1.25$ or M1 for 125% associated with 2450
13(c)	15	3	M2 for $1.048^n > 2$ or trial M1 for $9600 \times 1.048^n = 19200$